MAMMALS

A Zoology Study Guide to Class Mammalia



Figure 1. African elephants drinking at a river. Elephants are the largest living land mammals.

Introduction

Mammals are warm-blooded vertebrates belonging to the class **Mammalia**. There are more than 6,400 living species, ranging from the tiny bumblebee bat (about 2 grams) to the blue whale, the largest animal known to have ever lived. Mammals live on every continent and in every ocean, from polar ice to tropical rainforests and deserts.

This guide covers the defining features of mammals, their classification, body systems, reproduction, adaptations and conservation.

1. Characteristics of Mammals

Several features, found together, distinguish mammals from all other animal groups:

- **Mammary glands** – females produce milk to nourish their young.
- **Hair or fur** – present at some stage of life; helps insulation, camouflage and sensation.
- **Endothermy** – mammals maintain a constant, high body temperature.
- **Four-chambered heart** – completely separates oxygenated and deoxygenated blood.
- **Diaphragm** – a muscle that powers breathing through lungs.
- **Three middle-ear bones** – malleus, incus and stapes give acute hearing.
- **Heterodont teeth** – incisors, canines, premolars and molars, each with its own role.
- **Large, complex brain** – a well-developed neocortex supports learning and behaviour.

2. Classification

Living mammals are divided into three major groups according to how they reproduce: monotremes, marsupials and placentals.

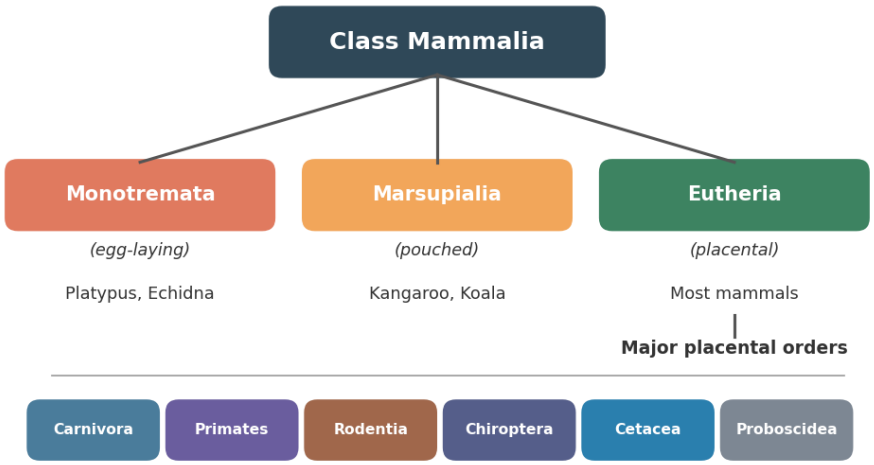


Diagram. Simplified classification of class Mammalia.

3. The Three Groups of Mammals

Group	Reproduction	Examples	Distribution
Monotremes	Lay leathery eggs; young are fed milk	Platypus, echidnas	Australia, New Guinea
Marsupials	Short pregnancy; young develop in a pouch	Kangaroo, koala, opossum	Australia, Americas
Placentals	Young nourished in the womb by a placenta	Elephant, whale, human, bat	Worldwide

Monotremes (Prototheria)

Only five living species exist: the platypus and four species of echidna. They lay eggs yet feed their young with milk, which is secreted from skin patches because they have no nipples.

Marsupials (Metatheria)

Marsupial young are born very small and undeveloped. The newborn crawls into the mother's pouch (marsupium), attaches to a teat and continues growing there. A joey kangaroo may remain in the pouch for around eight months.



Figure 3. A koala, an Australian marsupial.

Placentals (Eutheria)

The largest group, with more than 90% of all mammal species. The placenta lets the foetus develop for a long time inside the mother. Placental mammals are divided into about 19 orders, including those listed below.

4. Major Orders of Placental Mammals

Order	Key Features	Examples
Carnivora	Meat-eating; sharp canines and carnassial teeth	Lion, tiger, wolf, bear, seal
Primates	Grasping hands, forward-facing eyes, large brain	Monkeys, apes, humans
Rodentia	Continuously growing incisors; largest order	Rat, squirrel, beaver
Chiroptera	Only mammals capable of true flight (bats)	Fruit bats, insect bats
Cetacea	Fully aquatic; blowhole; flippers	Whales, dolphins, porpoises
Proboscidea	Trunk, tusks, very large body	Asian and African elephants
Artiodactyla	Even-toed hoofed mammals; many ruminants	Cow, deer, giraffe, pig
Perissodactyla	Odd-toed hoofed mammals	Horse, rhinoceros, tapir

5. Adaptations to Habitat

Mammals show remarkable adaptive radiation. Their limbs and bodies have been modified for different ways of life:

- **Running (cursorial):** horses and antelopes have long limbs and reduced toes.
- **Climbing (arboreal):** monkeys have grasping hands and often prehensile tails.
- **Burrowing (fossorial):** moles have strong, shovel-like forelimbs.
- **Swimming (aquatic):** whales and seals have streamlined bodies and flippers.
- **Flying (volant):** bats have wings formed from elongated finger bones covered by skin.

6. Mammals of the Sea: Cetaceans



Figure 4. A grey whale surfacing close to a boat.



Figure 5. An orca (killer whale), the largest dolphin.

Whales, dolphins and porpoises evolved from land-dwelling, hoofed ancestors about 50 million years ago. Although they live in water, they breathe air through lungs, give birth to live young and nurse them with milk. A thick layer of fat called **blubber** provides insulation and energy storage.

Cetaceans are divided into two groups: **baleen whales** (Mysticeti), which filter krill and small fish through plates of baleen, and **toothed whales** (Odontoceti), such as dolphins, orcas and sperm whales, which hunt actively and use **echolocation**.

7. Mammals of the Land: Primates and Hoofed Mammals



Figure 6. A mandrill, the largest monkey (a primate).

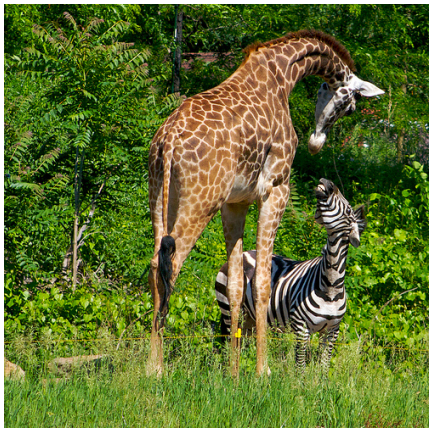


Figure 7. A giraffe, the tallest living mammal.

Primates have grasping hands, forward-facing eyes and large brains, and many live in social groups. **Hoofed mammals** such as the giraffe walk on the tips of their toes and are mostly plant-eaters. The giraffe is a ruminant with a very long neck supported by only seven neck bones.

8. Reproduction and Parental Care

Mammals typically invest heavily in a small number of offspring. After birth, young depend on milk and on parental care, during which they learn survival skills through play, imitation and instruction. Gestation periods vary greatly: about 12 days in the opossum, 280 days in humans and up to 22 months in the elephant, the longest of any mammal.

9. Digestive Adaptations

Diet strongly shapes the digestive system and teeth of mammals:

- **Herbivores** (cow, deer, elephant) have broad grinding molars and long intestines. Ruminants have a four-chambered stomach to digest cellulose.
- **Carnivores** (lion, wolf) have sharp canines and carnassial teeth with a short, simple gut.
- **Omnivores** (bear, pig, human) have a mixture of tooth types.
- **Insectivores** (hedgehog, shrew, anteater) have small, pointed teeth or none, and long sticky tongues.

10. Interesting Mammal Facts

- The **blue whale's** heart can weigh as much as 180 kg.
- The **platypus** detects prey using electroreception in its bill.
- The **giraffe** has only seven neck vertebrae, the same number as humans.
- The **pygmy shrew** is among the smallest mammals and must eat almost constantly.
- **Elephants** have a highly developed memory and strong social bonds.

11. Conservation

Roughly one in four mammal species is threatened with extinction according to the IUCN Red List. The main threats are habitat loss, hunting and poaching, climate change, pollution and the introduction of invasive species. Protection through national parks, anti-poaching laws, captive breeding and habitat restoration has helped species such as the Indian rhinoceros and the giant panda recover.

12. Summary

- Mammals are vertebrates with hair, mammary glands, a diaphragm and a four-chambered heart.
- The three groups are monotremes, marsupials and placentals.
- Placentals dominate, with orders such as Carnivora, Primates, Rodentia, Chiroptera and Cetacea.
- Adaptations allow mammals to run, climb, burrow, swim and fly.
- Many species need urgent conservation action.

Quick Review Questions

1. List four features that distinguish mammals from other vertebrates.
2. How do marsupials differ from placental mammals in reproduction?
3. Name the only order of mammals that can truly fly.
4. What is echolocation, and which mammals use it?
5. What role does blubber play in whales?